Robin Macharg

Senior / Principal iOS & Multi-Platform Software Engineer

+44 (0)7377 719820 - robin.macharg@gmail.com

LinkedIn | GitHub | StackOverflow

Professional Summary

Senior iOS & Security Architect: *Hands-on engineer with extensive experience developing secure, high-performance native iOS applications in Swift and Objective-C, combined with previous UK Government SC1 Security Clearance and secure device integration expertise.*

App Store Shipping & Scale: *Proven history of shipping and managing live applications on the Apple App Store, including sole ownership of a 13-app white-label portfolio driving £30M+ in annual revenue for blue-chip clients.*

Low-Level & Diagnostics Specialist: *Strong capability in C/C++ native integrations, memory optimization, low-level networking, and deep-dive crash log diagnostics developed while maintaining core enterprise SDK libraries.*

Continuous Automation & Delivery: *Highly proficient in modern devops and release tooling, including Xcode, TestFlight, App Store Connect, Fastlane, and GitHub Actions, successfully slashing release deployment times by up to 90%.*

Collaborative Engineering Leader: *Certified Scrum Master with a strong track record of mentoring international teams, establishing rigorous Git workflows, and successfully managing security audits and penetration testing.*

Alignment with Native iOS, Security & High-Performance Engineering Requirements

- **Native iOS Mastery (Swift & Objective-C):** Over a decade of deep native iOS experience designing clean, maintainable MVVM and Redux/TCA architectures using the iOS SDK, Xcode, and modern language paradigms.
- **App Store Shipping Experience:** Robust track record of publishing and maintaining enterprise-grade, high-traffic consumer and business applications live on the App Store using TestFlight and App Store Connect.
- **Performance, Security & Diagnostics:** Expert diagnostic skills utilizing Objective-C/C++ and low-level stack-trace analysis to resolve complex security, performance bottlenecks, and crash bugs at scale.
- **Secure Architectures & TEE/VPN Experience:** Direct experience building secure, custom VPN-encrypted applications for government projects, implementing enterprise SSO, and handling penetration testing remediations.
- **Automated Release Workflows:** Proficient in designing automated Fastlane and GitHub Actions pipelines to streamline App Store Connect distribution, code signing, and SDK delivery.
- **Collaborative Delivery & Git Workflows:** Certified Scrum Master with extensive experience leading Agile engineering teams, designing git-push hooks, and collaborating via structured peer-review pipelines.

Education, Certifications & Credentials

- **BSc (Hons) Artificial Intelligence and Computer Science (Joint)** — University of Edinburgh
- **Certified Scrum Master (CSM)** — Scrum Alliance
- Previous UK Government SC1 Security Clearance (6 concurrent DBS checks)

Selected Key Roles

Lead Mobile Developer | Boostworks Ltd.

November 2023 - Present (AppStore)

Keywords: [React Native, TypeScript, iOS/Android, Cross-Platform Architecture, AI-Assisted Engineering, CI/CD (Fastlane/Azure), OpenAPI, Apple/Google Pay, SSO, White-Label Scaling, Penetration Testing]

- Sole maintainer of a 60K-line multi-tenant codebase driving £30M+ in annual revenue (70% of company turnover) across 13 white-label iOS/Android applications for flagship enterprise clients including Costa, BMW, EasyJet, Heathrow, and GWR.
- Reduced mobile engineering costs by over 50% by successfully insourcing the entire 13-app catalog from an external agency and leading architectural migrations.
- Built end-to-end automation pipelines using Fastlane, Azure Pipelines, and custom TUIs, slashing new app creation and App Store deployment times by 90%.

- Led native component wrapping (Swift/Kotlin) for Apple Pay and Google Pay integrations, executed security penetration testing remediations, and integrated enterprise SSO.

Senior Developer | Floww Ltd.

May 2023 - September 2023 (AppStore)

Keywords: [iOS, SwiftUI, TCA, Redux, GitHub Actions, CI/CD, Mentoring, Shell Scripting, Optimization]

- Developed critical features for a high-security private investment market mobile platform using modern Redux-like Composable Architecture (TCA) and SwiftUI.
- Maintained and optimized CI/CD pipelines utilizing GitHub Actions and custom shell scripts to ensure robust code quality controls.
- Introduced and enforced development best practices, including SwiftUI previews, compile-time monitoring, git-push lint hooks, and App startup-time optimizations.

Senior Apple Engineer | Bugsnag Inc.

January 2020 - June 2020

Keywords: [Objective-C, C++, iOS, macOS, watchOS, tvOS, Open Source, Crash Reporting, Diagnostics, Optimization]

- Maintained and optimized open-source Bugsnag crash-reporting libraries for multi-platform Apple ecosystems serving major global enterprises like Netflix, Nintendo, Airbnb, and Amazon.
- Handled 3rd line support isolating complex native stack traces, low-overhead C/C++ exceptions, and publishing robust, lightweight diagnostic patch releases.
- Managed automated SDK packaging and multi-channel distribution across CocoaPods, Carthage, Swift Package Manager (SPM), and manual integrations.

Principal iOS Developer | Amiosec Ltd. (Contract)

March 2019 - December 2019

Keywords: [Xamarin, Visual Studio, C#, Objective-C/Swift Interop, EWS, ReactiveUI, Security, VPN, Charles Proxy]

- Architected and engineered a custom VPN-secured Exchange email and calendar client for the UK Government's high-security Foxhound project.
- Built high-performance, cross-language native interoperability between Xamarin/C# and Objective-C/Swift using .NET Embedding frameworks.
- Reverse-engineered Exchange Web Services (EWS) using Charles Proxy SSL Man-in-the-Middle decryption to construct a resilient local syncing mechanism.

Principal iOS Developer | Dyson Ltd. (Contract)

June 2017 - January 2020

Keywords: [iOS, Greenfield, Enterprise MDM, ARKit, RealmDB, Swift, Unit Testing, Penetration Testing]

- Sole developer delivering Dyson's first B2B sales iPad application ('Hero') from greenfield to global field test in 8 weeks, expanding the app to 23k lines of code.
- Achieved >80% code coverage across unit and UI test suites while utilizing custom Xcode templates to enforce a clean, testable MVVM architecture.
- Managed end-to-end enterprise Mobile Device Management (MDM via AirWatch) deployment, resolved penetration testing vulnerability findings, and integrated secure CMS backends.

Earlier Career Highlights

1996 - 2023 Multi-platform engineering, senior mobile SDK delivery, network analytics, and DevOps positions across diverse sectors.

- **Lead Mobile Developer** | *This One Ltd.* (SwiftUI, Combine, Async/Await, Python, Flask, Fastlane, GCP, Firebase, ML)
- **Senior iOS Developer** | *Sky* (Swift, Kotlin Multiplatform, SwiftUI, Video Streaming, Analytics, Agile)
- **Senior Developer** | *Living Map Ltd.* (iOS, Mapping SDK, XCFrameworks, CocoaPods, Carthage, Background Location, UX)
- **Head of Mobile Development** | *Actual Experience Ltd.* (Objective-C, Swift, CoreData, Realm, Networking, Jenkins, Vagrant)
- **Principal Mobile Developer (Contract)** | *Accident Exchange Group* (iOS, CoreData, CoreLocation, MapKit, RestKit)
- **Senior Middleware & DevOps Developer** | *MySpace* (Java, Python, JBoss, Drools, Puppet, Maven, RPM)
- **Senior Developer** | *RMS Services Ltd.* (Python, PostgreSQL, RESTful APIs, Asset Discovery)
- **Aerodynamics & Mass Properties Engineer** | *Airbus UK Ltd.* (Python, PyQt, Perl, Tk, PDL, OpenGL)
- **Server & Embedded Developer** | *Sift Ltd. / The Appliance Studio / Obtree / Circle.com / Glaxo Wellcome* (Linux, Java, Perl, Embedded Systems, Web Infrastructure)

Open Source Contributions

- Bugsnag Cocoa & C++: Core maintainer of open-source Apple platform crash-reporting libraries, resolving complex stack traces and low-level C++ exceptions.

- React Native Input Spinner: Key open-source contributor, developing and maintaining custom UI components for the cross-platform community.
- Pycel / Excel Formula Parser: Core contributor to this Python-based compiler mapping Excel dependency graphs and parsing formulas.
- Bricksmith: SourceForge award-winning LEGO CAD software for macOS, written in Objective-C and C.

Technical Skills and Expertise

Native iOS & Core Tooling - Swift, Objective-C, iOS SDK, Xcode, TestFlight, App Store Connect, Swift Package Manager (SPM), CocoaPods, Carthage, xcodebuild

Security, Performance & Diagnostics - UK SC1 Security Clearance, TEE/VPN-secured integrations, Penetration testing remediations, Crash reporting and diagnostics, Memory and compilation optimization, Background Location operations, Raw socket low-level networking

Architecture & Programming Paradigms - Clean Architecture, MVVM, Redux/TCA (Composable Architecture), MVC, ReactiveUI, White-Label Scaling, Kotlin Multiplatform Mobile (KMM)

Automation & DevOps Pipelines - Fastlane, Azure Pipelines, GitHub Actions, Jenkins, Custom Python/Shell Scripting, Git workflow automation (Probot/Octokit)

Secondary Languages & Databases - Python (FastAPI, Flask), C/C++, C#, Java, PostgreSQL, RealmDB, CoreData, OpenAPI/RESTful API design